

bdd Contributed Examples (contrib/)

This directory provides practical, ready-to-run examples demonstrating how to use `bdd` as both a **CLI tool** and a **native library (C-ABI and Python)** for real-world binary formats.

```
contrib/
├─ Makefile                # Builds C binaries and executes all tests
├─ README.md              # This documentation
├─ README.pdf             # Print-ready vector PDF documentation
├─ data/                  # Realistic binary sample files
│  └─ sample.mp3          # MPEG-1 Layer III audio frames (128k & 160k)
│  └─ sample.ts           # MPEG-2 Transport Stream (188B packets with PIDs 0, 256, 257)
│  └─ sample.wav          # RIFF WAVE 16-bit stereo PCM audio (44.1 kHz)
│  └─ sample_24bit.raw    # Raw 24-bit studio signed PCM audio
│  └─ sample.mp4          # ISO-BMFF MP4 with ftyp, mdat (H.264 NALs), and moov
│  └─ sample.jpg          # JPEG JFIF with SOF0 (640x480, 4:2:0 subsampling)
│  └─ sample.safetensors  # Safetensors file with FP8, BF16, and FP16 tensors
│  └─ sample_nvfp4.bin    # NVIDIA Blackwell NVFP4 (E2M1) 4-bit packed weights
│  └─ sample_fp6.bin      # OCP Microscaling FP6 (E3M2) 6-bit unaligned weights
│  └─ sample_ipv4.bin     # RFC 791 IPv4 20-byte base packet header
│  └─ sample_udp.bin      # RFC 768 UDP 8-byte datagram header
│  └─ sample_tcp.bin      # RFC 793 TCP 20-byte base segment header (SYN)
│  └─ sample_packets.bin  # Multi-packet capture stream (DNS/UDP, TCP SYN, HTTP GET)
├─ shell/                 # Standalone shell scripts using the bdd CLI
│  └─ decode_mp3.sh       # Unaligned 32-bit MP3 frame header parsing & frame seeking
│  └─ decode_mpeg_ts.sh   # MPEG-TS 13-bit PID inspection & container striding
│  └─ decode_wav.sh       # WAV header parsing, stereo demuxing & 24-bit downsampling
│  └─ decode_mp4.sh       # MP4 box traversal & H.264 NAL bitfield extraction (1U2U5U)
│  └─ decode_jpeg.sh      # JPEG SOF0 geometry & 4-bit chroma nibble extraction (4U4U)
│  └─ decode_ai_weights.sh # Safetensors 0(1) seeking & NVFP4/FP6/FP8/BF16 decoding
│  └─ decode_network.sh   # IPv4, UDP, and TCP header dissection with sub-byte flags
│  └─ bdd_kernel_inspect.sh # Bit-level inspection of Linux kernel structures (auxv, pagemap, PCI)
│  └─ stream_memory_keys.sh # Stream system raw memory 256 bits at a time and probe crypto keys
│  └─ run_all_shell_examples.sh # Master script running all shell demonstrations
├─ python/                # Python scripts using libbdd ctypes bindings
│  └─ generate_samples.py  # Deterministic generator for all data/ sample files
│  └─ decode_media.py     # MP3, MPEG-TS, JPEG, and H.264 NAL parsing
│  └─ decode_ai_weights.py # Safetensors inspection, NVFP4, and FP6 decoding
│  └─ decode_network.py   # IPv4, UDP, and TCP packet dissection and IP formatting
│  └─ bdd_ps.py           # Process status inspector slicing /proc/[pid]/pagemap, auxv & signal masks
│  └─ bdd_top.py          # Interactive top-like process monitor using bit-sliced metrics
│  └─ bdd_netlink_proc.py # Linux Netlink process connector socket monitor
│  └─ stream_memory_keys.py # Stream raw memory (physical, kernel, PID) and probe crypto keys
├─ c/                     # Native C programs linking against libbdd.so
│  └─ decode_media.c      # bdd_unpack_u64, bdd_pack_u64, and bit reversal
│  └─ decode_ai_weights.c # Native FP4, FP6, FP8, BF16, and FP16 float decoders
│  └─ decode_network.c    # High-performance IPv4/UDP/TCP unpacking with discrete flags
```

Quick Start

Run All Demonstrations

From the repository root:

```
make contrib
```

Or from inside `contrib/`:

```
cd contrib
make test
```

Regenerate Sample Data Files

To recreate all test binary files from scratch:

```
python3 contrib/python/generate_samples.py
```

Format & Bitstream Reference

1. MPEG-1 Layer III Audio (MP3) Frame Headers

Standard Unix tools cannot inspect MP3 frame headers because fields do not align to byte boundaries:

- **Header Bitfield (32 bits = 4 bytes):** 11U (Syncword 0x7FF) 2U (MPEG Version) 2U (Layer) 1U (CRC Protection) 4U (Bitrate Index) 2U (Sample Rate Index) 1U (Padding) 1U (Private) 2U (Channel Mode) 2U (Mode Extension) 1U (Copyright) 1U (Original) 2U (Emphasis)
- **CLI Pattern:**

```
bdd "11U2U2U1U4U2U1U1U2U2U1U1U2U" --count=1 --output-json < sample.mp3
```

- **O(1) Frame Seeking:**

```
# Jump directly to frame 2 at byte 417:  
bdd "417B:32 → 32" "11U2U2U1U4U2U1U1U2U2U1U1U2U" --count=1 --output-json < sample.mp3
```

2. MPEG-2 Transport Stream (MPEG-TS)

MPEG-TS uses repeating 188-byte containers (1504 bits). The 4-byte header contains an unaligned **13-bit Packet Identifier (PID)**:

- **Header Pattern:** Unpack the 32-bit header from each 188-byte packet while discarding the 184-byte payload automatically:

```
bdd "188B[0:32] → 32" 8U1U1U1U13U2U2U4U --count=4 --output-json < sample.ts
```

- **Direct Container PID Extraction:** Extract only the 13-bit PID (bits 11..24) from every 188-byte container with zero manual bitmasking or post-gap math:

```
bdd "188B[11:13] → 13" --output-integers < sample.ts
```

3. WAV / RIFF Audio & Multi-Channel Demuxing

- **Stereo Demuxing:** Skip the 44-byte RIFF header, frame into 32-bit stereo pairs (two 16-bit signed PCM samples), and demux into two independent mono audio files in a single pass (redirect stdout to `/dev/null` to discard the primary pass-through stream):

```
bdd "44B:32 → 32" 16S16S --input-little-endian \  
--demux-files="left.raw,right.raw" < sample.wav > /dev/null
```

- **Playing Raw Demuxed Channels with FFmpeg:** Raw PCM streams lack container headers, so specify sample format (`-f s16le`), rate (`-ar 44100`), and channel count (`-ac 1`):

```
# Play left and right channels:  
ffplay -f s16le -ar 44100 -ac 1 -autoexit left.raw  
ffplay -f s16le -ar 44100 -ac 1 -autoexit right.raw  
  
# Or convert directly to standard playable WAV files:  
ffmpeg -f s16le -ar 44100 -ac 1 -i left.raw left.wav  
ffmpeg -f s16le -ar 44100 -ac 1 -i right.raw right.wav
```

- **24-bit PCM Downsampling via Container Slicing:** Directly slice the most significant 16 bits (`[0:16]`) of each 24-bit PCM sample without needing manual arithmetic shifts:

```
bdd "24[0:16] → 16" 16S --output-tuples < sample_24bit.raw
```

4. MP4 / ISO-BMFF & H.264 NAL Units

- **MP4 Box Walking:** Inspect nested 32-bit big-endian length and 4-byte FourCC ASCII box headers:

```
bdd 32U32C --count=1 --output-json < movie.mp4
# Or seek directly to box offset:
bdd "28B:64 → 64" 32U32C --count=1 --output-json < movie.mp4
```

- **H.264 NAL Unit Headers:** Decode 32U (Length), 1U (Forbidden Zero), 2U (NAL Reference IDC), 5U (NAL Unit Type: SPS=7, PPS=8, IDR=5, Slice=1):

```
bdd "36B:40 → 40" 32U1U2U5U --output-json < movie.mp4
```

5. JPEG SOF0 Chroma Subsampling Nibbles

In baseline JPEG (0xFFC0), the horizontal and vertical chroma subsampling factors (e.g., 2×2 for 4:2:0) are stored as **4-bit nibbles** (4U4U):

- **Stream Pattern with Repetition Multiplier:**

```
bdd "24B:120 → 120" "8U16U16U8U3*(8U4U4U8U)" --count=1 --output-json < sample.jpg
```

6. AI Model Weights (Safetensors, NVFP4, FP6, FP8, BF16)

bdd natively decodes sub-byte and modern floating-point formats:

CODE	FORMAT	BIT WIDTH	DESCRIPTION
4E / 4e	NVFP4 (E2M1)	4 bits	NVIDIA Blackwell / OCP Microscaling FP4
6E / 6e	FP6 (E3M2)	6 bits	OCP Microscaling FP6
8E / 8e	FP8 (E4M3FN)	8 bits	NVIDIA Hopper H100 / OCP FP8
8Q / 8q	FP8 (E5M2)	8 bits	NVIDIA Ada Lovelace / OCP FP8
16Y / 16y	BF16	16 bits	Google Brain Bfloat16
16H / 16h	FP16	16 bits	IEEE 754 Half-Precision Float

- **Safetensors Inspection:**

```
# 1. Read 64-bit LE JSON header length:
HEADER_LEN=$(bdd "8*8U" --count=1 --output-json < model.safetensors | jq '.[0] + .[1]*256 + .[2]*65536 + .[3]*16777216')
# 2. Seek directly to tensor offset in O(1) time and unpack FP8 weights:
bdd "$(( 8 + HEADER_LEN ))B:8 → 8" 8E --output-json < model.safetensors
```

- **NVIDIA Blackwell NVFP4 (Sub-Byte 4E4E):**

```
bdd 4E4E --output-json < sample_nvfp4.bin
```

- **OCP FP6 (Unaligned 4*6E across 24 bits):**

```
bdd "4*6E" --output-json < sample_fp6.bin
```

7. Network Protocol Headers (IPv4, UDP, TCP)

Network headers are strictly big-endian (network byte order) with MSB-first bit numbering. bdd parses these bitfields with natural-order U and B specifiers, cleanly extracting sub-byte flags without manual bit shifting:

- **RFC 791 IPv4 Base Packet Header (160 bits = 20 bytes):**

- **Preset:** --preset=ipv4-header
- **Bitfield:**
version:4U,ihl:4U,dscp:6U,ecn:2U,total_length:16U,id:16U,flags:3U,frag_offset:13U,ttl:8U,protocol:8U,checksum:16U,src_ip:32U,dst_ip:32U
- **CLI Command:**

```
bdd --preset=ipv4-header --output-json --json-object < sample_ipv4.bin
```

- **RFC 768 UDP Datagram Header (64 bits = 8 bytes):**

- **Preset:** `--preset=udp-header`
- **Bitfield:** `src_port:16U,dst_port:16U,length:16U,checksum:16U`
- **CLI Command:**

```
bdd --preset=udp-header --output-json --json-object < sample_udp.bin
```

- **RFC 793 TCP Base Segment Header (160 bits = 20 bytes):**

- **Preset:** `--preset=tcp-header`
- **Bitfield:**
`src_port:16U,dst_port:16U,seq_num:32U,ack_num:32U,data_offset:4U,reserved:3U,ns:1B,cwr:1B,ece:1B,urg:1B,ack:1B,psh:1B,rst:1B,syn:1B,fin:1B`
- **Sub-byte Flags:** Every control flag (`ns` , `cwr` , `ece` , `urg` , `ack` , `psh` , `rst` , `syn` , `fin`) is isolated into an individual boolean/integer field.
- **CLI Command:**

```
bdd --preset=tcp-header --output-json --json-object < sample_tcp.bin
```

- **Multi-Packet Capture Stream Dissection:** Combine stream offset patterns (`<offset>B:<size> → <size>`) with presets or named fields:

```
# Packet 1 (DNS over UDP at offset 0):
bdd --preset=ipv4-header --count=1 --output-json < sample_packets.bin
bdd "20B:64 → 64" --preset=udp-header --count=1 --output-json < sample_packets.bin
bdd "28B:32 → 32" "dns_id:16U,flags:16U" --count=1 --output-json --json-object < sample_packets.bin

# Packet 2 (TCP SYN at offset 32B):
bdd "32B:160 → 160" --preset=ipv4-header --count=1 --output-json < sample_packets.bin
bdd "52B:160 → 160" --preset=tcp-header --count=1 --output-json < sample_packets.bin
```

8. System Raw Memory Key Probing (256-bit Unit Streaming)

When investigating memory dumps, physical RAM, or running processes for cryptographic material, cryptographic keys (such as AES-256, ChaCha20, Ed25519, and SHA-256 state) appear as contiguous 256-bit blocks with maximum Shannon entropy (≈ 5.0 bits/byte) and balanced bit distributions ($\approx 50\%$ ones set).

The contributed tools (`contrib/shell/stream_memory_keys.sh` and `contrib/python/stream_memory_keys.py`) stream system memory in discrete 256-bit units (`--input-unit=256` , 32 bytes per unit) and run `bdd --probe-keys=256` to locate candidate keys:

- **Supported Sources:**

- `demo` : In-memory realistic stream with embedded NIST AES-256 and ChaCha20 test vectors (runs without root).
- `kcore` : Linux kernel virtual address space via `/proc/kcore` .
- `devmem` : Physical RAM via `/dev/mem` (requires root/sudo).
- `self` : Current process address space via `/proc/self/mem` (runs without root).
- `pid <PID>` : Target daemon or process memory via `/proc/<PID>/mem` .
- `<FILE>` : Raw memory dump or disk image file.

- **Streaming and Probing Example (Shell):**

```
# Run realistic in-memory test demo (128 units = 4096 bytes):
./contrib/shell/stream_memory_keys.sh --source demo --units 128

# Output as machine-readable JSON:
./contrib/shell/stream_memory_keys.sh --source demo --units 128 --json

# Probe 4096 units (128 KiB) of kernel memory:
./contrib/shell/stream_memory_keys.sh --source kcore --units 4096

# Inspect a specific process:
./contrib/shell/stream_memory_keys.sh --source pid 1234 --units 2048
```

- **Streaming and Probing Example (Python):**

```
python3 contrib/python/stream_memory_keys.py --source demo --units 128
python3 contrib/python/stream_memory_keys.py --source self --units 1024
```

- **Manual One-Liner Pipeline:**

```
# Stream 4096 blocks of 32 bytes (256 bits each) directly into bdd:
dd if=/proc/kcore bs=32 count=4096 status=none | \
    bdd --input-unit=256 --probe-keys=256 --output-json
```

9. Linux Kernel Structures & Hardware Telemetry (/proc & /sys)

Linux exposes raw kernel binary structures, MMU page translation tables, and bus controller registries via virtual filesystems (/proc and /sys). Traditional shell utilities cannot unpack bitfields like 64-bit pagemap entries (which contain individual 1-bit flags for presence, swap, dirty state, exclusive ownership, and 55-bit Physical Frame Numbers) or 16-byte ELF auxiliary vectors without compiling custom C programs.

`bdd` provides native presets and $O(1)$ hardware seeking to slice these structures directly from the command line or Python:

- **Virtual Memory Page Table Slicing (/proc/[pid]/pagemap):**

- **Preset:** `--preset=proc-pagemap`
- **Bitfield:** `present:1b,swapped:1b,file_page:1b,3x,uffd_wp:1b,exclusive:1b,soft_dirty:1b,pfn:55u`
- **Addressing & Seeking:** Each 4096-byte virtual page corresponds to one 64-bit entry in `pagemap`. Slicing any virtual address requires seeking to bit offset $(VADDR / 4096) * 64$:

```
# Inspect 4 pages of the stack for PID 1234:
bdd --input-file="/proc/1234/pagemap" --input-skip-bits=OFFSET_BITS \
    --preset=proc-pagemap --count=4 --output-json --json-object
```

- **Memory Metrics:**

- `present=1, exclusive=1`: Unique Set Size (USS / private physical memory).
- `soft_dirty=1`: Page modified since last clear.
- `swapped=1`: Page swapped out to disk (PFN field contains swap type and swap offset).

- **ELF Auxiliary Vector Traversal (/proc/[pid]/auxv):**

- **Preset:** `--preset=proc-auxv`
- **Record Structure:** 16-byte pairs of `Elf64_auxv_t { uint64_t a_type; uint64_t a_val; }`.
- **Bitfield:** `val:64U,type:64U` with `--input-little-endian`
- **Key Variables:**
 - `AT_CLKTCK` (type 17): Kernel clock ticks per second (used for exact CPU percentage calculations).
 - `AT_PAGESZ` (type 6): System MMU page size (typically 4096 bytes).
 - `AT_SECURE` (type 23): Setuid/privileged security execution flag.
 - `AT_ENTRY` (type 9): ELF entry point address.

- **CLI Command:**

```
bdd --input-file="/proc/self/auxv" --preset=proc-auxv --output-json --json-object
```

- **PCI Hardware Configuration Space (/sys/bus/pci/devices/*/config):**

- **Preset:** `--preset=pci-config`
- **Bitfield:** `bist:8U,hdr_type:8U,latency:8U,cache_line:8U,class_code:8U,subclass:8U,prog_if:8U,rev_id:8U,status:16U,cmd:16U,device_id:16U,vendor_id:16U`
- **Hardware Telemetry:** Unpacks Vendor ID (e.g., `0x10de` NVIDIA, `0x8086` Intel), Device ID, Command/Status register flags, and PCI Class code without requiring external utilities like `lspci`.
- **CLI Command:**

```
bdd --input-file="/sys/bus/pci/devices/0000:00:00.0/config" \
    --preset=pci-config --output-json --json-object
```

- **Process Deep-Inspection Tool (contrib/python/bdd_ps.py):** Dissects `/proc/[pid]/pagemap` to calculate true Unique Set Size (USS), dirty pages, and shared pages, parses `/proc/[pid]/auxv` for timer clock ticks and SUID status, and decodes `/proc/[pid]/status` 64-bit signal masks (`SigPnd`, `SigBlk`, `SigIgn`, `SigCgt`) into human-readable POSIX signal names (`INT`, `QUIT`, `TERM`, `WINCH`):

```
# Deep-inspect current running processes:
python3 contrib/python/bdd_ps.py

# Target a specific PID:
python3 contrib/python/bdd_ps.py --pid 1234

# Sort by Unique Set Size (USS / private memory):
python3 contrib/python/bdd_ps.py --sort uss

# Emit machine-readable JSON:
python3 contrib/python/bdd_ps.py --json
```

- **Interactive System & Process Monitor (`contrib/python/bdd_top.py`)**: Full-screen terminal dashboard displaying real-time per-core CPU utilization meters, system RAM/Swap gauges, process thread counts, and bit-sliced USS memory metrics:

```
# Launch interactive top monitor (refresh every 1.5s):
python3 contrib/python/bdd_top.py

# Set custom refresh interval:
python3 contrib/python/bdd_top.py --interval 0.5
```

Interactive Hotkeys:

- `c` : Sort by CPU utilization (%)
- `m` : Sort by Resident Set Size (RSS)
- `u` : Sort by Unique Set Size (USS)
- `p` : Sort by Process ID (PID)
- `q` : Quit

10. Real-Time Process Lifecycle Event Streaming (`NETLINK_CONNECTOR`)

Polling `/proc` to detect process creation or termination introduces CPU overhead, drains mobile battery, and misses ephemeral, short-lived processes (such as rapid compiler invocations, transient scripts, or malicious injection).

Linux provides the **Netlink Process Connector** (`NETLINK_CONNECTOR = 11` , `CN_IDX_PROC = 1` , `CN_VAL_PROC = 1`), an event-driven multicast socket that streams binary notifications directly from the kernel scheduler whenever a process event occurs.

- **Header Preset (`--preset=netlink-proc-event`)**: Unpacks the 16-byte `proc_event` header from the Netlink message payload:

```
timestamp_ns:64U,cpu:32U,what:32U with --input-little-endian
```

- **Process Event Types:**

- `FORK` (`0x0001`): Parent PID/TGID creates child PID/TGID.
- `EXEC` (`0x0002`): Process executes a new binary image (resolves new comm and command line).
- `EXIT` (`0x80000000`): Process terminates, emitting exit code, exit signal, and lifetime duration.
- `UID` (`0x0004`) / `GID` (`0x0040`): Privilege transitions (e.g. `setuid` , `sudo` , dropping capabilities).
- `COMM` (`0x0200`): Thread renaming (e.g. Rayon, Tokio, or Go worker pool threads).

- **Zero-Drop High-Throughput Burst Handling**: During heavy build workloads (e.g. `cargo build -j16`), the kernel can emit thousands of fork/exec/exit events per second, causing standard Netlink sockets to fail with `ENOBUFS` (Errno 105, buffer space exhausted). `contrib/python/bdd_netlink_proc.py` eliminates drops through three kernel socket optimizations:

1. **8MB Socket Buffer**: Sets `SO_RCVBUF` to 8 MiB (`8 * 1024 * 1024`).
2. **`NETLINK_NO_ENOBUFS`**: Sets socket option `NETLINK_NO_ENOBUFS = 5` on protocol level `SOL_NETLINK = 270` , instructing the kernel to prioritize continuous streaming without raising fatal socket errors.
3. **Multi-Message Iteration**: Reads 64 KiB chunks and steps through concatenated `nlsghdr` records using `NLMSG_ALIGN` / `NLMSG_NEXT` pointer arithmetic.

- **Running the Monitor:**

```
# Stream all process events in real time:
sudo python3 contrib/python/bdd_netlink_proc.py

# Filter specific lifecycle events (e.g., EXEC or FORK):
sudo python3 contrib/python/bdd_netlink_proc.py --event EXEC

# Stream machine-readable NDJSON for security telemetry / SIEM pipelines:
sudo python3 contrib/python/bdd_netlink_proc.py --json
```

Native C & Python APIs

C API (`include/bdd.h` & `libbdd.so`)

```
#include "bdd.h"

// 1. Unpack unaligned bitfields (e.g. UDP 64-bit header)
uint64_t udp_fields[4];
bdd_unpack_u64("16U16U16U16U", udp_u64, udp_fields, 4);
// fields: [src_port, dst_port, length, checksum]

// 2. Unpack TCP Word 3 flags and window size in one step
uint64_t tcp_word3[12];
bdd_unpack_u64("4U3U1B1B1B1B1B1B1B16U", w3, tcp_word3, 12);
uint64_t syn_flag = tcp_word3[9]; // Isolated SYN bit!

// 3. Bit-exact repacking
uint64_t repacked = 0;
bdd_pack_u64("16U16U16U16U", udp_fields, 4, &repacked);

// 4. Convert AI floating point numbers
double w0 = bdd_decode_fp4_e2m1(0x03); // 1.50
double fp8 = bdd_decode_fp8_e4m3(0x38); // 1.00
double bf16 = bdd_decode_bf16(0x3F80); // 1.00
```

Python API (`python/bdd.py`)

```
from bdd import Bdd

b = Bdd()

# 1. Unpack network datagram headers
src_port, dst_port, length, checksum = b.unpack("16U16U16U16U", udp_u64)

# 2. Extract TCP 32-bit word 3 with discrete control flags
offset, res, ns, cwr, ece, urg, ack, psh, rst, syn, fin, win = b.unpack(
    "4U3U1B1B1B1B1B1B1B16U", tcp_w3
)

# 3. Bitfield roundtrip pack
packed_u64 = b.pack("16U16U16U16U", (5353, 53, 12, 0x8899))

# 4. AI float codecs
val_fp4 = b.decode_fp4(0x03)
val_fp8 = b.decode_fp8(0x38)
val_fp6 = b.decode_fp6(0x10)
```